

OFFICIAL TECHNICAL DOCUMENTATION

MCA WALLAH

Full-Stack Portfolio & CMS Engine

A comprehensive architectural blueprint, component design, database model specifications, REST API reference, and administrative maintenance manual.

AUTHOR / CREATOR

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PROJECT VERSION

1.0.0 (Production Release)

TECHNOLOGY STACK

**React 19, Vite, Node.js, Express,
MongoDB**

TARGET AUDIENCE

**Software Engineers, Tech Recruiters &
Clients**

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1. Executive Summary & Purpose

1.1 Project Vision

The **MCA WALLAH Portfolio** is an enterprise-grade personal branding platform engineered to demonstrate high-level full-stack development expertise. Unlike static portfolio templates, this application is a dynamic, living web system powered by an autonomous RESTful backend, persistent MongoDB database, and a full-featured administrative studio (CMS).

1.2 Core Business & Technical Objectives

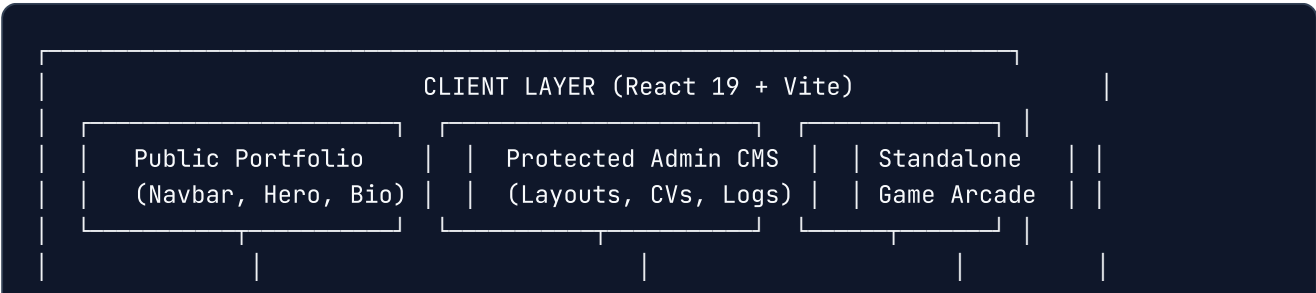
- **Dynamic Content Customization:** Eliminate hardcoded content by enabling real-time administrative modifications for every visual section (Hero, Navbar, Resume Vault, Skills, Projects, and Games).
- **Modern Visual Excellence:** Deliver state-of-the-art UI/UX using custom CSS design tokens, frosted glassmorphism (`backdrop-filter: blur(28px)`), and dynamic responsive layout switching.
- **Interactive Engagement:** Maximize visitor retention through an interactive Retro Gaming Arcade (Snake, Unbeatable Minimax AI Tic-Tac-Toe, Typing Challenge, 15-Puzzle) and a movable Sticky Background Engine.
- **Permanent Resume Library:** Maintain a secure repository of candidate CVs and resumes with dynamic button toggling and multi-resume inline preview/download capabilities.

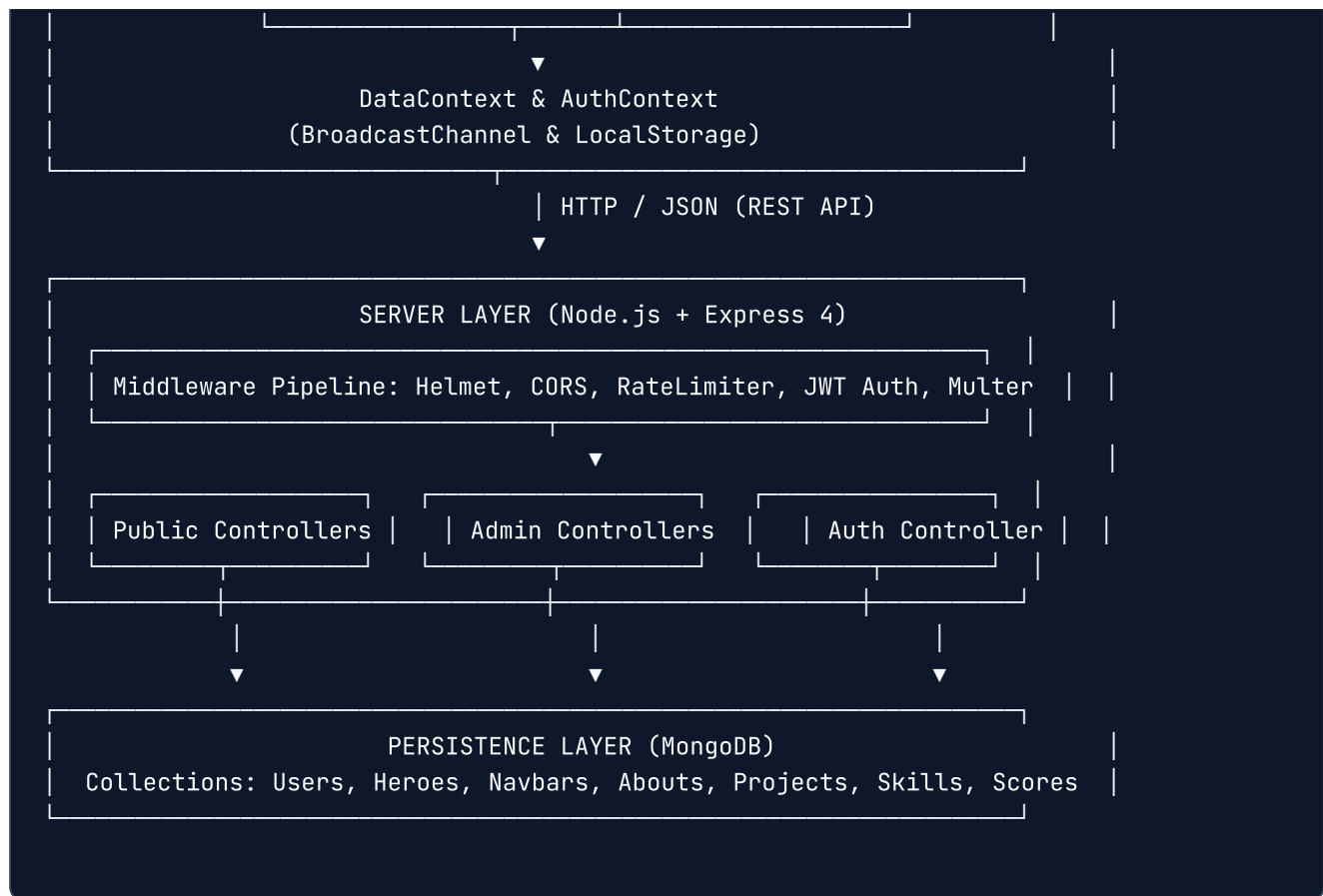
Key Innovation: Real-Time Multi-Tab Synchronization

The application employs the modern `BroadcastChannel` API alongside optimistic React state updates. Changes made in the Admin CMS reflect instantly on open portfolio visitor tabs without requiring full page reloads.

2. System Architecture & High-Level Design

2.1 High-Level Architecture Diagram





3. Frontend Architecture & Design Systems

3.1 Dynamic Navigation Engine

The navigation bar is built with an adaptive auto-fit geometry (`width: max-content; max-width: min(96vw, 1320px)`) that prevents element clipping and supports 4 distinct presentation presets configurable from the Admin panel:

Preset ID	Design Aesthetic	Visual Attributes
<code>floating-dock</code>	macOS Frosted Glass Island	Central capsule, <code>backdrop-filter: blur(28px)</code> , subtle shadow aura
<code>cyber-capsule</code>	Cyberpunk Neo-HUD	High-contrast dark base, glowing neon border, cyan/magenta accents
<code>minimal-island</code>	Ultra-Clean Compact Pill	Streamlined geometry, auto-expanding pill boundary, tight padding
<code>full-width</code>	Classic Enterprise Sticky Bar	Edge-to-edge full width with bottom border and glass background

3.2 Multi-Style Hero Section

The Hero section accommodates diverse candidate identities through 4 structural layout engines:

- 💎 **Glassmorphic Floating Capsule:** Centered glassmorphic card with dynamic stat chips and call-to-action buttons.
- ⚡ **Two-Column Developer Split:** Left-hand copy paired with a live interactive macOS Code Terminal with syntax highlighting and 1-click clipboard copying.
- 🎨 **Minimalist Editorial Spotlight:** Bold high-impact typography with subtle radial ambient backdrops.
- 🕹️ **Retro-Futuristic Hologram:** Angular cyberpunk badges, technical coordinates, and animated scanlines.

3.3 Draggable Floating Backdrop Widget (`⦿ Background`)

Visitors can freely reposition the theme customization widget anywhere on the desktop or mobile viewport via dual mouse/touch event listeners. Coordinates are saved in `localStorage` so the preferred widget position persists across browser visits.

4. Permanent Multi-Resume Vault Engine

The resume subsystem solves the limitation of single-file static PDF links:

- **Permanent Cloud Vault:** Every resume uploaded via Admin Studio is permanently archived in the backend storage with filename, file size, and upload timestamp.
- **Multi-Button Visibility Controller:** The administrator can dynamically toggle visibility for up to **3 concurrent resume buttons** (e.g. *Full Stack CV*, *Frontend Resume*, *Academic Bio*).
- **Glassmorphic Dropdown & Inline Preview:** If multiple resumes are active, the header seamlessly transforms into an interactive dropdown with direct download () and new-tab preview () capabilities.

5. Interactive Gaming Arcade & Minimax AI

5.1 Arcade Games Overview

Game Title	Type & Mechanism	Algorithmic / Technical Highlight
 Retro Snake	Grid-based Canvas Game	Tick-rate velocity scaling, collision detection, and score multiplier.
 Tic-Tac-Toe AI	Strategy Board Game	Unbeatable Minimax Algorithm with recursive alpha-beta tree evaluation.
 Speed Typing	Developer WPM Challenge	Character-by-character accuracy parser, real-time WPM calculation, timer hook.
 15-Puzzle Sliding	Permutation Grid Puzzle	Solvability inversion checker, swap matrix, move counter.

5.2 Dual Window Mode (Inline & Standalone Window)

Games can be played inside the interactive modal or launched into an isolated, distraction-free standalone window at `/arcade`. High scores are automatically synced to the MongoDB global leaderboard.

6. Protected Admin Studio CMS Architecture

6.1 Administrative Modules Overview

Panel Route	Module Name	Administrative Capabilities
<code>/admin/dashboard</code>	 Analytics Overview	Real-time visitor counts, message analytics, and system status indicators.
<code>/admin/navbar</code>	 Navbar Studio	Switch layout presets, customize brand prefix/suffix, style "Let's Talk" CTA.
<code>/admin/hero</code>	 Hero & Header CMS	Change hero layout modes, edit typewriter roles, modify CTA anchors and pills.
<code>/admin/about</code>	 Profile & Resume Vault	Upload permanent resumes, manage button visibility limits, edit bio and stats.

Panel Route	Module Name	Administrative Capabilities
/admin/projects	 Projects Manager	Create/Edit projects, assign tech tags, upload preview cards, manage order.
/admin/skills	 Skills Matrix	Add skill icons, assign categories (Frontend/Backend/DevOps), set proficiency %.
/admin/games	 Arcade & Leaderboard	Manage active games, descriptions, and moderate leaderboard high scores.
/admin/messages	 Feedback Inbox	Read, mark as read, and delete contact form inquiries.
/admin/settings	 Account & Credentials	Change Admin username, email, password, and export JSON backups.

7. Database Schemas & Data Modeling

7.1 Core Mongoose Schemas Summary

```
// 1. User & Admin Schema
const UserSchema = new Schema({
  name: { type: String, required: true },
  email: { type: String, required: true, unique: true, lowercase: true },
  password: { type: String, required: true }, // BCrypt Hashed
  role: { type: String, enum: ['user', 'admin'], default: 'user' },
  avatar: { type: String },
  preferences: {
    darkMode: { type: Boolean, default: true },
    background: { type: String, default: 'mesh' },
    accentColor: { type: String, default: '#e84545' }
  }
}, { timestamps: true });

// 2. Navbar Schema
const NavbarSchema = new Schema({
  layoutStyle: { type: String, default: 'floating-dock' },
  logoText: { type: String, default: 'Mahadeb' },
  logoPrefix: { type: String, default: '<' },
  logoSuffix: { type: String, default: '>' },
  showHireMeButton: { type: Boolean, default: true },
  hireMeButtonText: { type: String, default: "Let's Talk" },
  hireMeStyle: { type: String, default: 'gradient-glow' },
  hireMeIcon: { type: String, default: 'fa-solid fa-paper-plane' },
  hireMeTarget: { type: String, default: 'contact' }
}, { timestamps: true });

// 3. About & Resume Vault Schema
const AboutSchema = new Schema({
  bio: [String],
  resumes: [{
    title: String,
    url: String,
    isPrimary: Boolean,
    activeOnPortfolio: Boolean,
    fileSize: String,
    uploadedAt: { type: Date, default: Date.now }
  }]
});
```

8. RESTful API Endpoint Documentation

Method	Endpoint Route	Access Level	Description
GET	/api/portfolio/public	Public	Aggregates all active portfolio data in a single high-performance payload.
POST	/api/portfolio/contact	Public	Validates feedback message and sends automated Nodemailer notification.
POST	/api/auth/login	Public	Authenticates credentials and returns JWT Access & Refresh tokens.
PUT	/api/user/profile	Protected (JWT)	Updates authenticated admin's display name and login email.
PUT	/api/user/password	Protected (JWT)	Verifies current password and updates to new bcrypt-hashed password.
POST	/api/user/reset-defaults	Protected (JWT)	Restores factory default credentials (mahadeb@portfolio.com).
POST	/api/admin/section/:type	Admin Guard	Updates CMS section items in MongoDB and broadcasts cross-tab updates.

9. Security, Authentication & Rate Limiting

9.1 Authentication Architecture

- **JWT Dual-Token Protocol:** Short-lived Access Tokens (15m) paired with rolling Refresh Tokens (7d) stored securely in client storage.
- **BCrypt Hashing:** Passwords salted with 10 rounds before database persistence.
- **Route Guards:** Protected React Router wrapper (`AdminAuthGuard.jsx`) validates role privileges before rendering administrative panels.

9.2 API Protection & Headers

- **Helmet Security Headers:** Configures XSS filters, CSP protections, and prevents iframe clickjacking.
- **Express Rate Limiting:** Prevents DDoS and brute-force attempts on sensitive endpoints like `/api/auth/login` and `/api/portfolio/contact`.

10. Deployment, Production & Troubleshooting Guide

10.1 Vercel SPA Routing Configuration (`vercel.json`)

To prevent `404: NOT_FOUND` errors on client-side routes (`/admin/login` , `/admin/dashboard` , `/arcade`), the root project contains:

```
{
  "rewrites": [
    {
      "source": "/(.*)",
      "destination": "/index.html"
    }
  ]
}
```

10.2 Default Administrative Credentials Quick Reference

Parameter	Default Production Value
Admin Login Portal	<code>http://localhost:5173/admin/login</code>

Parameter	Default Production Value
Default Email	<code>mahadeb@portfolio.com</code>
Default Password	<code>Admin@123456</code>
Reset Mechanism	Admin Studio $\rightarrow$ Settings $\rightarrow$ "Reset to Default Credentials"